Automotive Electric Propulsion Systems With Reduced or No Permanent Magnets: An Overview

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***Abstract*—Hybrid and electric vehicle technology has seen rapid development in recent years. The motor and the generator are at the heart of the vehicle drive and energy system and often utilize expensive rare-earth permanent magnet (PM) material. This paper reviews and addresses the research work that has been carried out to reduce the amount of rare-earth material that is used while maintaining the high efficiency and performance that rare-earth PM machines offer. These new machines can use either less rare-earth PM material, weaker ferrite magnets, or no magnets; and they need to meet the high performance that the more usual interior PM synchronous motor with sintered neodymium–iron–boron magnets provides. These machines can take the form of PM-assisted synchronous reluctance machines, induction machines, switched reluctance machines, wound rotor synchronous machines (claw pole or biaxially excited), double-saliency machines with ac or dc stator current control, or brushless dc multiple-phase reluctance machines.**

***Index Terms*—Automotive drives, motors, permanent magnets (PMs).**

I. Introduction

With ever increasing environmental and fuel energy concerns, the focus on automotive electrification has dramatically increased in the last decade. The propulsion system of the vehicle has received much attention, together with the associated power electronics and energy storage devices (batteries, supercapacitors, and fuel cells). Existing commercial hybrid electric vehicles (HEVs) and electric vehicles (EVs), such as Toyota Prius, Honda Insight, Chevrolet Volt, Nissan Leaf, and several other vehicles from manufacturers such as Renault and BMW, all use high-energy-density permanent magnet (PM) machines for electric propulsion. The magnetic material is usually sintered neodymium–iron–boron (NdFeB). Recent sharp increases and variations in the price of NdFeB magnets to more than 150 US$/kg are due to a virtual market monopoly. There is also potential for rapid depletion of rare-earth magnet resources because of the future extensive use of the material in wind generators, automotive drives, and home appliance application. Addressing these issues is a challenge, and use of less rare-earth magnet material is a good solution.

In this paper, the reduction in the amount of rare-earth magnet material in the drive motor and generator is addressed. This is a theme that many manufacturers are working toward. There are production vehicles such as the Tesla sports cars that do not use PM motors. Before discussing recent motor and generator developments, it is important to briefly discuss energy storage and drive train arrangements.

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II. HEV and EV Drive Systems

A. Energy Storage

Automobile electrification is more than a century old. EV development was restricted by energy storage limitations. However, recent developments in new batteries, such as lithium ion batteries, have expanded the potential for EVs. These batteries offer longer range and higher power density. Typically, they can be in the energy range of 70–200 Wh/kg and power range of 400–600 W/kg. The battery pack life cycle has been extended to over five years, whereas the cost has reduced [2]. Fig. 1 compares various energy storage technologies.

B. Converter Efficiency and Vehicle Hybridization

The introduction of the insulated-gate bipolar transistor (IGBT) inverter and the development of the Sic-IGBT inverter have led to rapid advancements in HEV and EV technology. This is due to increased inverter efficiency (around 95% for IGBT inverters and 97%–98% efficiency for Sic-IGBT inverters), lower volume (10 kVA/L), and reasonable cost (although still above 20 US$/kVA). The hybridization factor for a vehicle can be defined as [2], [3]

$$\mathrm{HF} = \frac{P_{\mathrm{electric}}}{P_{\mathrm{electric}} + P_{\mathrm{ICE}}} \tag{1}$$

where P_{electric} is the electric drive power, and P_{ICE} is the internal combustion engine (ICE) power. Four practical HEV systems are illustrated in Fig. 2. These are microhybrids, mild

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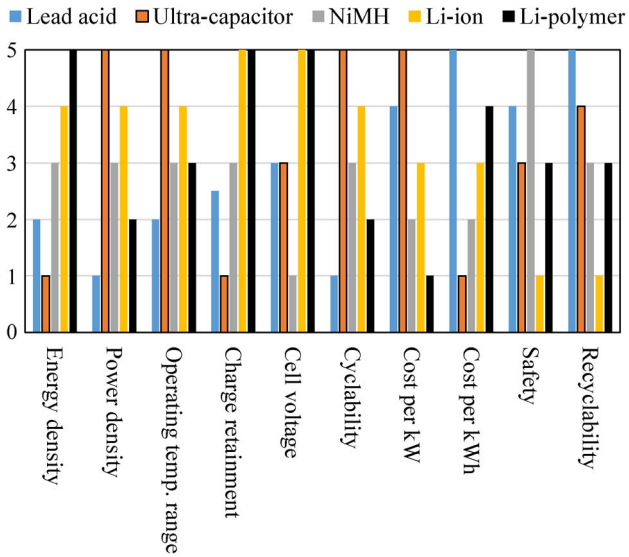

Fig. 1. Battery and ultracapacitor attributes when applied to HEVs and EVs [2]—arbitrary scale from 1 to 5.

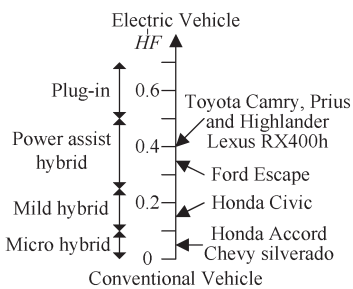

Fig. 2. HEV classification by hybridization factor HF [2].

hybrids, power-assist hybrid, and plug-in hybrids [2]. Obviously, a vehicle with an HF of 1 is an EV.

C. HEV Drive Train Topologies and Power Split Devices

Several HEV drive trains have been developed; however, the parallel configuration has become the standard commercial HEV drive train arrangement. Fig. 3(a) shows the Toyota Prius configuration in the Toyota Hybrid Synergy II version (2004) [4]. One of the key components is the power split device [Fig. 3(b) shows the 1997 Toyota electric variable transmission (EVT) [5]]. The ICE, motor, and generator will all transmit different and independent power at different speeds during the driving cycle, and the power split device offers flexibility to the transmission.

Further developments have been aimed at combining the motor and the generator with the gearing to remove the mechanical planetary gearbox and combine the motor and the generator. This leads to two electric machines in one frame and two pulsewidth-modulated (PWM) bidirectional converters with a power split device; this is shown in Fig. 4 [5], [6]. One disadvantage is that one "brushless" PM machine actually has a spinning stator winding with a bulky brush gear that requires maintenance. Automotive machines tend to be low-voltage high-current machines; hence, the brushes will have to conduct very high currents. The power split device allows

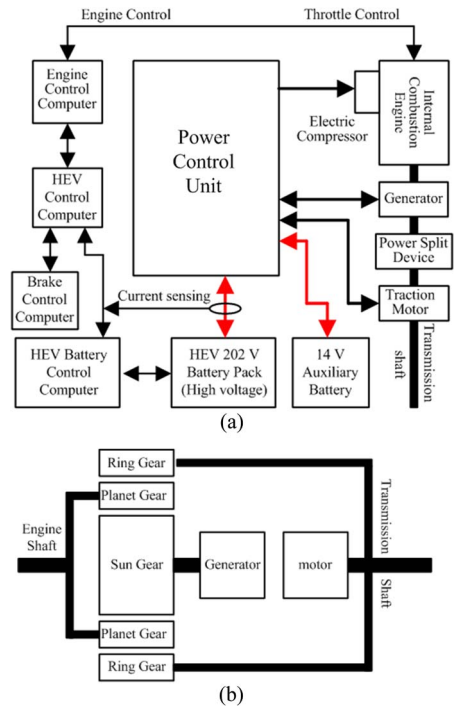

Fig. 3. Parallel Toyota Prius HEV power split device, also known as planetary gear EVT. (a) Toyota Prius parallel hybrid system—Toyota Hybrid Synergy II [4]. (b) Toyota Prius power split device [5].

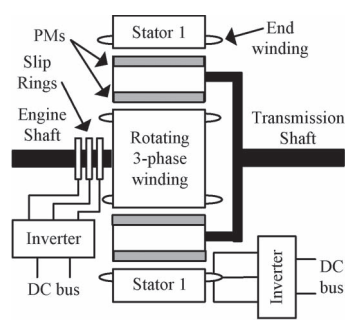

Fig. 4. Integration of two PM motors into the EVT [5], [6] (one brushless machine now has a rotating stator winding; thus, slip rings are required).

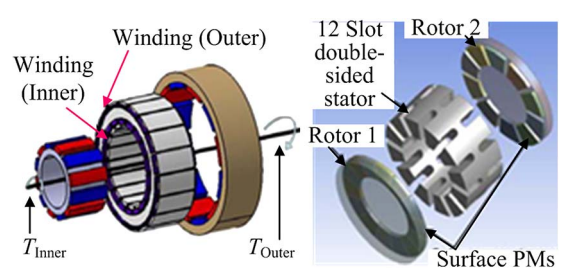

Fig. 5. Brushless PM EVTs [7], [8]. Radial flux and axial flux topologies (reproduced with permission from the authors).

the ICE to operate near its "sweet spot" on the torque/speed curve for maximum efficiency and minimum air pollution. In the integrated EVT in Fig. 4, the power split device uses high-energy PM synchronous machines (PMSMs). The brushes can be removed; and radial and axial flux examples of integrated brushless machines that could be utilized in the EVT are shown in Fig. 5 [7], [8]. These machines still use NdFeB magnets.

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TABLE I
TOYOTA AND EXPERIMENTAL MOTOR COMPARISON

Parameter [mm unless otherwise stated]	Air cooled	Liquid Cooled	
	Experimental IPMSM [11]	IPMSM Prius 2nd Generation [12]	IPMSM Prius 3rd Generation [13]
Outside diameter	146	269	264
Stack length	224	84	50
Active volume [L]	3.75	4.78	2.74
Winding type	Distributed	Distributed	Distributed
Pole number	8	8	8
Slot fill factor	0.6	0.84	0.84
Air-gap length	0.70	0.73	0.73
Speed range [rpm]	5000-12000	1200-6000	2790 – 13900
Power [kW]	35	50	50
Torque [Nm]	70	400	207
Current density [A/mm^2]	6	18	19
Torque density per active volume [Nm/L]	18.7	83.8	75.6
Knee point eff. [%]	90	85	88

D. Current Technology

The presence of two electrical machines (see Fig. 3—a motor and a generator) with distributed windings and strong PMs (either salient pole or surface magnet rotor) means that there is some degree of standardization in production machines. Table I [9], [10] gives some examples of the second- and third-generation Toyota Prius and an experimental machine. The saliency present in interior PMSMs (IPMSMs) provides an additional reluctance torque, which can significantly improve the performance [12]. Alternatively, Honda has used a machine with 18 slots, 16 poles and tooth wound concentrated windings [14]. This machine has no reluctance torque, and surface magnets can be used. The third-generation Prius generator also employs this sort of arrangement with 12 stator slots.

The development of the IPMSM drive in HEVs and EVs has typically led to machines that can produce 400 $\mathrm{N \cdot m}$ at 18-A/mm^2 conductor current density in the winding and 83-$\mathrm{N \cdot m/L}$ torque density up to a base speed of 1500 r/min. At 1500 r/min, the overall system efficiency is 85% (the motor electromagnetic efficiency is around 95%—the system has additional inverter, battery, and mechanical losses), and the machine can produce 50 kW up to 6000 r/min—this is for the second-generation Toyota Prius and is often used as a benchmark. These drives tend to have a narrow constant torque range and an extended constant power range (often termed the field weakening region). The later third generation of Toyota Prius drive produces 200 $\mathrm{N \cdot m}$ up to 2800 r/min at 19 $\mathrm{A/mm^2}$ and 75-$\mathrm{N \cdot m/L}$ torque density, with 89% overall efficiency at 2800 r/min, and 50 kW of power from 2800 up to 13 900 r/min. While the torque and the amount of magnet material are almost half, the speed is doubled to produce the same power. Therefore, an exercise in reducing the magnet material was carried out by doubling the speed and halving the torque. For a torque density of 45 $\mathrm{N \cdot m/L}$, the efficiency at the knee point may rise to 95% (slightly lower at higher speeds due to increased copper losses). There is substantial phase advance in these machines to obtain high reluctance torque [12].

E. Magnetless Motor Review

Competitive electric motors that do not use NeFeB magnets have been the subject of recent investigations. These machines are discussed in the following sections:

- Section III: induction machines (IMs) with copper cage rotors [9], [10], [12], [15];
- Section IV: switched reluctance motors (SRMs) with standard tooth-wound three-phase windings [16]–[19] and diametrical-coil double-stator segmented-rotor four-phase windings [20];
- Section V-A: ferrite PM-assisted synchronous reluctance machines [21], [22] with PM demagnetization and a novel biaxial excitation generator for automobiles (BEGA) [23];
- Section V-B: claw pole machine without or with ferrite interior PM (IPM) in the rotor [24]–[26];
- Section V-C: brushless dc (BLDC) multiple-phase reluctance machines (BLDC-MRM) [27]–[29];
- Section V-D: double-saliency multiple-phase reluctance machine with dc–ac stator windings [30], [31];
- Section V-E: ferrite-stator-PM dual-saliency switched-flux machines [32] or ferrite rotor dual-saliency switched-flux machines [33];
- Section V-F: the standard dc-excited synchronous machine [34], [35].

As already stated, a discussion is put forward for these machine topologies within the context of HEVs and EVs. Some are utilized commercially, whereas others are still at the research and development stage or even the early research stage. There is commercial interest in these types of drive motor if they can reduce the cost of the materials; rare-earth material is sourced from only a few countries, and the price and supply can be volatile. The perception is that rare-earth brushless PM motors offer very high torque density ([36] quoted Mekhiche, Program Director for hybrid drive solutions, BAE Systems, U.K.: "Rare-earth magnets allow for lighter motors with higher torque—between 30% and 60% less weight than equivalent induction motors, with an increase in efficiency of 5% or more."). A commercial review of the different drive systems was given in [37], although here, the machines are addressed from a technical viewpoint; [38] also reviewed some of the basic automotive drive and motor topologies used. They quoted power density and efficiency in several types of motors as follows: brushed dc 2.5, IM 3.5, IPMSM 5, and SRM 3.5 on an arbitrary scale up to 5.

III. COPPER CAGE IM

The IM has been utilized in HEV and EVs [9], [10], [12], and some interesting variations have been reported in [15]. They have been used in commercial electric cars such as the Telsa sports car [39], although a notable example of an early adaptation of this drive was General Motors (GM) EV1 [40]. A recent retrofit conversion (the cars are virtually rebuilt) has been on the DeLorean sports car from the early 1980s [41]. It is interesting to note the use of IMs in high-performance sports cars. Eberhard, the former Technical Director of Tesla Sports Cars, Inc., stated [43]: "With a gasoline engine, performance

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comes with a big penalty—if you want a car that has the ability to accelerate quickly, you need a HP engine, and you will get poor gas mileage even when you are not driving it hard. On the other hand, doubling the horsepower of an electric motor improves efficiency. It is therefore quite easy to build an electric car that is both highly efficient and very quick." This suggests that electric drives are very applicable to high-performance vehicles.

In [38], vehicles that utilize an IM drive are listed: Renault Kangoo, Chevrolet Silverado, Daimler Chrysler Durango, and BMW 5. The obvious advantages to this type of machine are that they are cheap, robust, and have no PMs. As noted in [38], they are not as efficient (with high rotor losses) or as torque dense as a PM machine, the inverter usage factor can be low, and the control (usually a flux vector control) at low speed can be an issue. Much work has been done to improve the controllability [42].

Issues with efficiency will affect the range and the required battery capacity. The IM magnetization current leads to higher stator copper losses and large peak kilovoltampere in the converter that supplies the IM. Care should be exercised in IM thermal design. In [12], a comparison was made between the Toyota Prius second-generation IPMSM and an example copper cage rotor IM and an example SRM. IMs for HEVs tend to be rated at about 50 kW and operate between 6000 and 12 000 r/min [12], [44]. A 50-kW motor may be suitable for a small EV; however, for a high-performance EV, the power may be much higher (the Tesla sports car is 225 kW in the range of 5000–6000 r/min). Since these machines are relatively small in volume, a cast copper rotor with low resistance and improved efficiency can be used.

Casting copper cages is a relatively new technology, and the economic case for them was put forward in [45]. They may appear to be the way forward, but alternative arrangements have been studied. In [46], a compound rotor was investigated that had an alloy outer cage with magnetic properties and an aluminum inner cage. This topology led to improved torque at standstill since the flux is increased and there is less flux concentration in the rotor tooth tips. Another comparison between an IM and a switched reluctance drive was published in [47]. To reduce losses even further, the study reported in [48] considered superconduction. If in-wheel drive motors are used, then the motor can be fabricated with an outer cage rotor [49], which will help increase the mean air-gap diameter and, hence, the torque.

One important aspect of many designs is the use of forced cooling, usually with water or a glycol/water mix. While [9] outlined consideration of forced cooling and used a commercial thermal analysis software package, others have also considered and tested the thermal performance of the IM [50], [51].

In terms of performance, Table II compares the Prius second-generation IPMSM, an example SRM, and an example IM [12]. The outer envelopes of the machines are the same, although it was necessary to vary the rotor diameter. For similar sizing, the IM has the median integral cost and the lowest efficiency at peak torque, but the efficiency is competitive at peak speed. The table also shows operation at the base speed of 1500 r/min and maximum torque and a maximum speed of 6000 r/min with a power of 50 kW. For the IM, the inverter requirements have to be considered. Even if the IM is designed with a low per unit (p.u.) short circuit reactance, the rated current and power at base speed cannot be maintained when the speed further increases. If the maximum inverter voltage cannot be more than 150% of the base speed voltage, then a wide constant power speed range (CPSR), greater than 2.5 times the constant torque range, is not possible because of reactive power increase. There are a few solutions to address this problem.

- Use two inverters for feeding the stator windings from both ends: one handles the active and the other the reactive power, to obtain a typical 4.5 CPSR [15]. This is at a cost of having more than 100% additional inverter capacity.
- Design the IM for a voltage V_{smax} at the base speed and maximum power, which is around 70% of the maximum available inverter voltage. This allows for 30% oversizing in inverter capacity.
- Use a bidirectional Z source converter to produce the 130% V_{smax} during the wide CPSR.
- Use twin inverters, which are 50%–60% of the rated kilovoltampere of the machine and connect them such that

TABLE II
COMPARISON OF IPMSM, IM, AND SRM [12]

Geometrical Parameters

Parameter	PM Motor	SRM	Induction Motor
Outer stator diameter [mm]	269.0	269.0	269.0
Inner rotor diameter [mm]	111.0	111.0	111.0
Outer rotor diameter [mm]	160.5	170.0	180.0
Axial core length [mm]	84	84	84
Weight of stator core [kg]	18.65	14.11	11.86
Weight of stator copper [kg]	5.99	7.44	10.57
Weight of rotor core [kg]	5.22	5.16	6.15
Weight of rotor magnet/copper [kg]	1.30	---	7.67
Total weight [kg]	31.16	26.71	36.25

Comparison of Performance at 1500 and 6000 RPM

Speed = 1500 rpm

Variable	Torque [Nm]	Current [Arms or Apk- SRM]	Iron loss [W]	Copper loss [W]	Eff. [%]	RMS current density [A/mm^2]
IPM	303	141.1	198	4328	91.3	15.7
IM	297	164.8	148	8591	83.1	15.8/12.1
SRM	294	300	404	7653	85.2	20.1

Speed = 6000 rpm

Variable	Torque [Nm]	Current [Arms or Apk-SRM]	Iron loss [W]	Copper loss [W]	Eff. [%]	RMS current density [A/mm^2]
IPM	45.6	31.8	953	219	96.1	3.75
IM	50.8	47.1	439	730	95.2	4.51/3.72
SRM	52.1	60	4074	306	88.2	4.02

Approximate Costs of Motors Excluding Frame and Fittings

		Laminated Steel	Copper	Neodymium Iron Boron	Totals [US$]
	US$/kg	1.3[a]	6.6[b]	132[c]	
IPM	Weight	23.87	5.99	1.3	
	Cost	31.03	39.53	171.60	242.17
SRM	Weight	19.27	7.44	0	
	Cost	25.05	49.10	0	74.16
IM	Weight	18.01	18.24	0	
	Cost	23.41	120.38	0	143.80

[a]Lamination estimated to be approximately double bulk steel cost.
[b]London Metal Exchange, June 2010
[c]W. T. Benecki, The Permanent Magnet Industry Outlook, Great Western Minerals Group, June 2008

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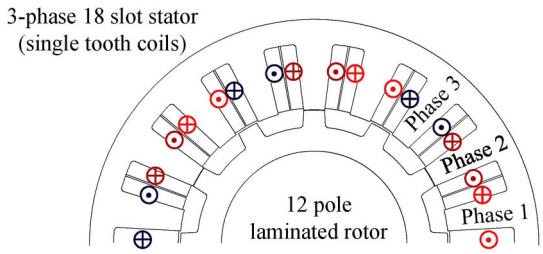

Fig. 6. Three-phase 18-slot 12-rotor-pole SRM [12] showing winding and geometry.

they allow for electronic pole changing. The pole number ratio is 2 [52].

The results in [12] show that it is difficult to obtain the same torque density as an IPMSM; thus, a less volume constrained application is needed. In an HEV, the space for the drive train is more restricted due to the presence of the ICE and the EVT; EVs tend to be more straightforward, and thus, they are more amenable to IM use. In [53], it is noted that IMs are suitable for hybrid light trucks, such as the Hino 4T Ranger HEV, and for buses, such as the GM Allison Hybrid system (parallel hybrid) and the BAE system HybriDrive. This paper also notes several other current systems in HEVs and EVs. One notable system is the GM two-mode hybrid. One advantage of the IM is that it can be quite flexible in phase number, and it is possible to control two induction motors from one inverter, and the phase number of these machines can be different [54].

IV. Switched Reluctance Machine

In early 2012, it was reported that, at the "EV Japan" show, 30% of the display vehicles had magnetless drive motors [55], but most were IMs. SRMs are now beginning to be introduced into vehicles. The Land Rover will utilize an SRM in 2014 [56], and Toyota appears to be in partnership with Renault–Nissan and is working on SRM drives [57]. Most SRMs are conventional in topology with a radial air-gap flux and double saliency, although axial air-gap flux is possible [58], and this is a double-sided axial flux, which can significantly increase torque density. For radial flux machines, the rotor is usually inside the machine [16]–[19], although it is possible to have a rotor external to the stator for in-wheel operation [59]; double-sided-rotor radial-flux operation is also a possibility [60].

For the conventional SRM, there are different numbers of salient poles on the stator and rotor. The current tends to be unipolar and trapezoidal, and the phase number can vary; mostly, three phases are used, but there are many occurrences of two-, four-, and five-phase SRMs. Using too many phases will complicate the inverter, and even numbers of poles lead to asymmetrical winding so that magnetic energy cannot be cycled from one phase to another during commutation if the switching is overlapped. This can affect efficiency. Fig. 6 shows a half cross section of a typical SRM, and this was the arrangement used in [12]. This was a theoretical study and gives an indication to what design considerations are required. Some results are shown in Fig. 7, where the efficiency decreases to 87% at 300 N · m and 1500 r/min. At 6000 r/min and 50 kW, the efficiency reaches a very good 95%.

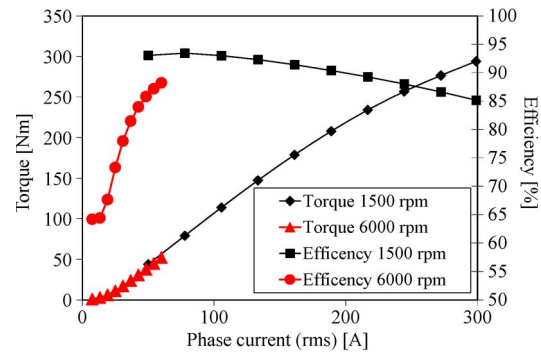

Fig. 7. Torque and efficiency at 1500 r/min (base speed) and 6000 r/min (maximum speed) for three-phase 18-slot 12-rotor-pole SRM [12].

TABLE III
Comparison of SRM Designs in [16] and [19]

Parameter or Performance	1st SRM [16]	2nd SRM [19]
Outer diameter [mm]	269	269
Stack length [mm]	135	135
Stack and coil-end length [mm]	160	154
Air-gap length [mm]	0.5	0.5
Iron core	10JNEX900	10JNEX900
Weight [kg]	39.6	39.7
Wire diameter [mm]	0.95	0.6
Wire turns	13	17
Parallel turns (strands in hand)	12	22
Slot fill factor [%]	54.1	57
Peak current [A]	380	320
Max current density [A/mm^2]	24	33
RMS current @ 1200 rpm [Arms]	240	206
Maximum torque [Nm]	340	400.4
Torque density [Nm/L]	38.3	45

A recent thorough investigation of an SRM [16] has been carried out on a 500-V dc 50-kW machine, which operated over a range up to 14 000 r/min. The case study was an attempt to match the performance of a Toyota third-generation IPMSM, which peaks at just over 200 N · m. The researchers managed to push the SRM performance up to 100 kW, which is higher than the IPMSM. However, it should be pointed out that the laminations are very thin (aeronautical motor specification—10JNEX900), which will increase material and manufacturing costs. The authors worked on improving the duty cycle requirements for an SRM [17], [18]. They carried out an extensive redesign in [19] in order to obtain similar torque to the Toyota second-generation motor. They presented further results for the machine in [16]. Simulation results from the theoretical redesign, some salient parameters, and performance figures are given in Table III. The machine under study was an 18/12 machine having 0.1-mm laminations with low core loss at high flux density (10INEX900) material. The machine was operated at a very high current density (33 A/mm^2) and could produce the required 400 N · m at 45 N · m/L. The predicted RMS current was 206 A at 1200 r/min, which is about the same as that for the NdFeB second-generation motor. The efficiency was also similar at about 90% at a peak torque of 400 N · m and 1200 r/min. However, at 50 kW and 6000 r/min, it is around 92%, which is lower than that of the second-generation motor (95%), but this is not substantial. What needs to be pointed out is that the current density is high and needs to be reduced (although this is a peak value of a unipolar near-trapezoidal

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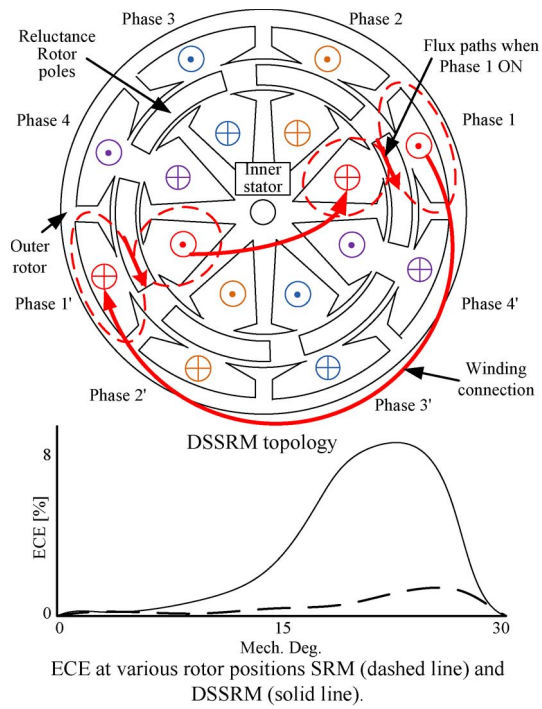

Fig. 8. DSSRM [20]. (Top) Topology and (bottom) illustration of reduction of ECE.

waveform). Even with a state-of-the-art thermal system, this is still too high. Generally, current densities up to about 25 A/mm^2 RMS are thought to be possible in an automotive drive motor, provided the machine is operated with a duty cycle and forced fluid cooling is used. The authors comment that this is what they will address in further work.

Another SRM topology is the double-stator switched reluctance machine (DSSRM) [20]. An example topology is shown in Fig. 8. The aim of this design is to reduce noise and radial forces. The machine consists of an inner and an outer stator with a rotor placed between them. The rotor and stators are subjected to radial forces that can cause noise and vibration. These are a function of loading; thus, energy conversion efficiency (ECE) is defined to quantify these radial forces, i.e.,

$$\mathrm{ECE}(\theta, i) = \frac{F_{\mathrm{motion}}(\theta, i)}{F_{\mathrm{motion}}(\theta, i) + F_{\mathrm{radial}}(\theta, i)} \times 100\% \quad (2)$$

where F_{motion} is the net force in the direction of rotation, and F_{radial} is the net radial force on the rotor. It is shown in Fig. 8 that the ECE is much reduced in the DSSRM, which will reduce the vibration and noise, as well as bearing wear.

The researchers then went further in [10], where a design exercise on the DSSRM was carried out (and compared with an IPMSM and an IM). Two designs were developed: an air-cooled 8/6 machine that could deliver 29 kW and a fluid-cooled 12/8 machine that could deliver 50 kW. Table IV lists some machine parameters and motor performance figures. The current densities are much lower than those in a standard SRM. This is a much more complex structure; thus, the cost of the DSSRM is expected to be 20%–30% more than that of a commercial SRM. The end winding will be also very long. In the example shown in Fig. 8, the coils have almost a full 180 mechanical degree coil pitch. However, it is shown that the length of the flux path is relatively short so that the inductance is low, which reduces the magnetization and reactive power VAr requirement of the inverter; possibly, it is only 15% higher than a similar rated IM. It should be mentioned that, even for an eight-pole IM, the power factor at peak torque and base speed may be quite low (< 0.6), which means poor peak kilovoltampere utilization.

TABLE IV
DESIGN PERFORMANCE [10]

Parameter or Performance	DSSRM-1 air-cooled	DSSRM-2 Fluid cooled
Outer diameter [mm]	288	210
Stack length [mm]	152.5	75
Pole number	8/6	12/8
Volume [L]	6.26	2.6
Winding type	Distributed	Distributed
Slot fill factor [%]	50	60
Speed range [rpm]	2000-6000	4000-10000
Power [kW]	29	50
Torque [Nm]	137	113
Air-gap length (each) [mm]	0.5	0.4
Current density [A/nn^2]	6	10
Torque density [Nm/L]	21.9	43.5
Efficiency at base speed [%]	88	93

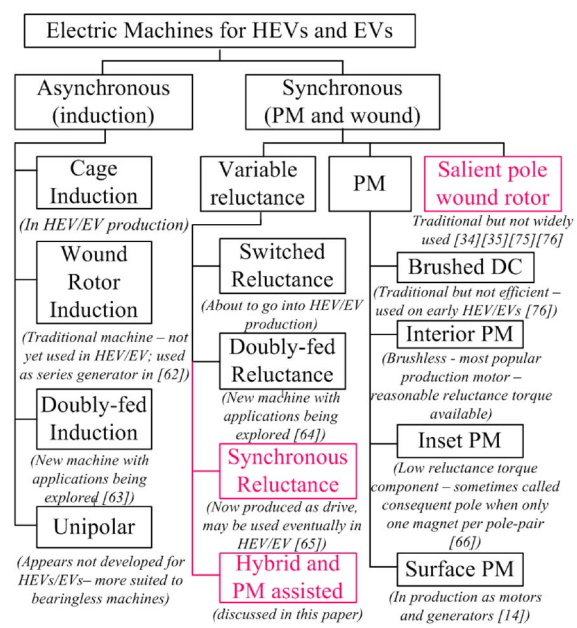

Fig. 9. Chart of possible motor choices for HEVs/EVs [61]. Additional alternatives are added in red.

V. NONPRODUCTION RESEARCH MACHINES

So far, the IM and the SRM have been discussed as magnetless alternative machines in HEVs and EVs. For the reduced-magnet or no-magnet motors, the IM is used in production, whereas the SRM is about to enter production. We will now review some more unusual machine topologies that are in early stages of research and development. Several possibilities were given in a chart in [61], and this is further developed in Fig. 9, where wound rotor synchronous, synchronous, and PM-assisted reluctance machines are added. The chart is annotated and gives some references to some of the machines that are being

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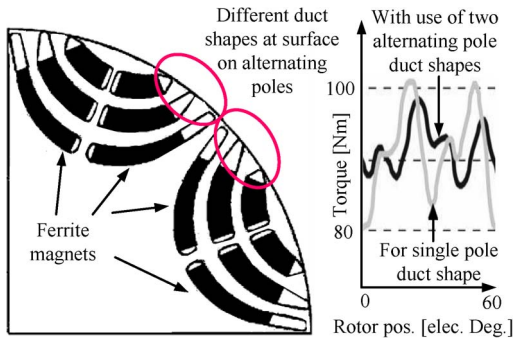

Fig. 10. Eight-pole PM-assisted synchronous reluctance machine with distributed stator winding—reluctance ducts (multilayer flux barriers) vary on alternate pole to reduce torque ripple [21] and contain magnets.

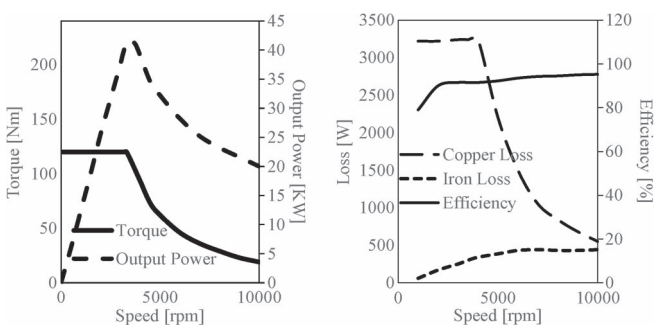

Fig. 11. Eight-pole PM-assisted synchronous reluctance machine—torque, power characteristics, and loss component [21].

developed. In the following sections are alternatives that have not yet reached production.

A. Ferrite-PM-Assisted and Field-Wound-Assisted Synchronous Reluctance Machine

Examples of the ferrite-PM-assisted synchronous reluctance machine were reported in [21] and [22]. In HEVs and EVs, the rare-earth IPMSMs used have Lq/Ld ratios, possibly in the range of 1.5–2.5. The second-generation Prius machine is capable of generating a reluctance torque almost equal to excitation torque [12]. Reluctance machines typically do not have high torque density; thus, one technique to address this problem is to use a ducted rotor with multiple flux barriers (see Fig. 10) and a flux barrier on the quadrature axis created using ferrite PMs. This combination will result in a high-saliency rotor with an Ld/Lq ratio in the region of 3–5 or more, which may lead to competitive torque density and efficiency [67]. This ratio is provided that the ferrite PMs do not demagnetize when there is a peak torque (high I_q) and the temperature is low [22].

An example was studied in detail in [21], and the rotor is illustrated in Fig. 10, with loss and efficiency characteristics shown in Fig. 11. The performance of this machine appears reasonable. One advantage of this topology is that, under no-load conditions and, effectively, over speeding, the voltage may be less than 1.5–1.7 p.u. so that voltage protection may be eliminated. This implies that the CPSR is wide so that it is 3–4, which is similar to HEV and EV IPMSMs. The bulk of the torque when is it peaking and running up to the base speed is produced by the reluctance effect. Using a phase advance

TABLE V
SPECIFICATIONS OF THE PROPOSED PM-ASSISTED SYNCHRONOUS RELUCTANCE MACHINE [21]

Parameter	Value
Pole number	8
Stator outer diameter [mm]	250
Rotor outer diameter [mm]	169.1
Stack length [mm]	40
Air-gap length [mm]	0.9
Max current density [A/mm^2]	20
Coercive force of PMs [kA/m]	11

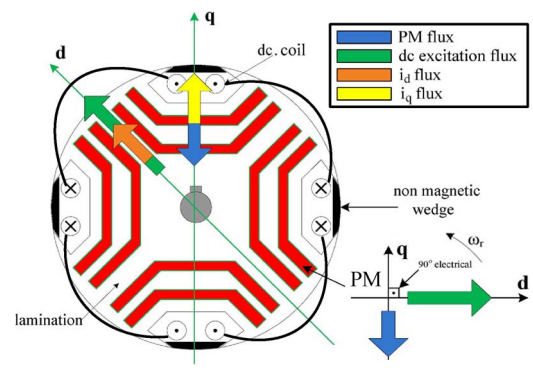

Fig. 12. BEGA machine topology showing the d-axis ducted rotor with ferrite magnets on the q-axis and dc field windings on the d-axis [23] (reproduced with permission from the authors).

technique together with the effects of q-axis magnet flux will allow the motor to run through an extended power range. Much of this work focused on reducing torque ripple. Smooth torque and low inertia are advantageous in HEV/EV drive motors. Smooth torque will reduce wear on mechanical components and tires; low inertia is required because the motors do not have a gearbox, and at high speed, the stored energy can make the driving uncomfortable (in a similar way to driving an ICE vehicle in a gear that is too low).

The results for the machine in [21] are encouraging; it appears that the machine can produce at 20 A/mm^2 about 75% of the torque of the Toyota second-generation IPMSM for the same size and with liquid cooling. Parameters for the machine are given in Table V.

The ferrite-PM-assisted synchronous reluctance machine may suffer from demagnetization at low temperature [22], which can be avoided by preheating before start. Careful shaping of the ferrite PM rotor can help. Although a 50-kW machine or an axially laminated anisotropic (large saliency) rotor machine has not been developed yet, the results suggest it should be investigated further.

One of the drawbacks of the ferrite-PM-assisted synchronous reluctance machine is its inability to maintain a flat power characteristic for a CPSR of 4. This has led to the idea of adding dc excitation windings on the d-axis via brush gear. The machine still retains the characteristic of L_d being much greater than L_q with ferrite PMs on the q-axis. The BEGA was proposed in [23], and a cross section of the rotor is shown in Fig. 12. The stator has a standard three-phase distributed winding. The voltage equation is

$$\bar{V}_s = R_s \bar{I}_s + \frac{d\overline{\Psi}_s}{dt} + j\omega_r \overline{\Psi}_s. \tag{3}$$

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The flux has several terms and can be written as

$$\overline{\Psi}_s = \Psi_d + j\Psi_q = L_d i_d + L_{mF} i_F + j(L_q i_q - \Psi_{PMq}). \quad (4)$$

The electromagnetic torque is then

$$T_e = \frac{3}{2} p_1 \left(L_{mF} i_F i_q + (L_d - L_q) i_d i_q + \Psi_{PMq} i_d \right). \quad (5)$$

When the current is on the q-axis and regulated so that the q-axis stator flux Ψ_q is approximately zero, the BEGA works at unity power factor. Under these conditions,

$$\Psi_s = \Psi_d = L_{md} i_F \qquad i_d = 0 \quad (6)$$

so that the torque is given by

$$T_e = \frac{3}{2} p_1 L_{md} i_F i_q = \frac{3}{2} p_1 L_{md} i_F \left(\frac{\Psi_{PMq}}{Lq} \right). \quad (7)$$

The power factor is unity when operating. The ideal CPRS $= \infty$ because the back electromotive force (EMF) E_q can be controlled by control of the dc excitation flux. Therefore, it is not a function of speed. In addition, the PM flux opposes the q-axis current; thus, in the phasor diagram, the $jX_q I_q$ voltage is offset by an EMF induced by the magnets. These are illustrated in (4). In reality, it is limited by mechanical losses. The flux/current interaction is maximum at 90° because the flux is solely produced by the field current I_f, which is fed from a four-quadrant dc–dc low-power converter via slip rings. The field current can change polarity when needed. The large stator magnetization current I_d is replaced by the dc field current; and the dc–dc converter kilovoltampere is less than 3% of the machine kilovoltampere rating. This arrangement at peak torque reduces the peak current by 30% if it is assumed to be a reluctance machine with $I_d = I_q$. This will result in a similar kilovoltampere reduction of the converter. The multiple-flux-barrier rotor is fragile, but critical speeds can be avoided by design. A larger 50- or 100-kW design for the BEGA has yet to be built and tested, but the 2-kW results [23] suggest that this may be worth investigating.

In [67], a different variation is described. This machine has radial reluctance laminations and homopolar flux with an axial field winding. However, it is magnetless and is subject to patenting and development with results that look promising. The machine is described as the "U motor."

B. Claw Pole Machine Without and With Ferrite IPMs Rotor

The claw pole machine has been used for several decades as a vehicle alternator. It has been developed for use as a combined motor/generator [24], [25], [68]. It has been also modified further into more unusual topologies. In [26], V-shaped ducted rotor pole pieces are used with orthogonal magnets (spoke magnets in a radial flux machine). In [69], two axial rotor segments are utilized, which are offset with no pole interleaving. These can use ferrite or a reduced amount of rare-earth magnet.

The claw pole alternator (CPA) has a diode-rectified output and rotor field current output control. A decade ago, it was used as a starter/alternator for the first 42-V dc mild HEV in

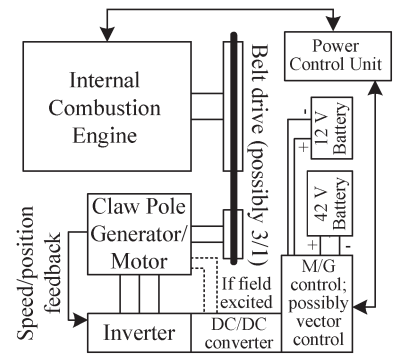

Fig. 13. Generic arrangement for claw pole motor/generator in a mild HEV system.

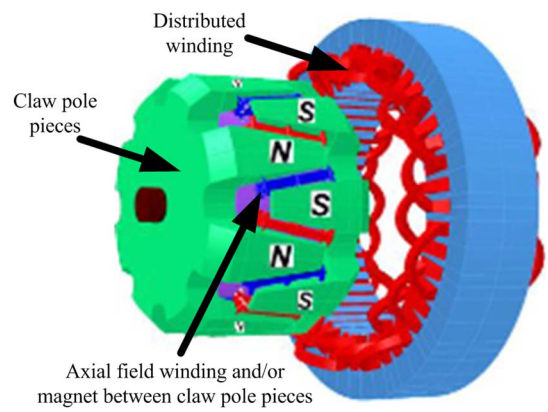

Fig. 14. Claw pole machine (taken from [24] with permission from the authors).

Toyota Crown Royal [68]. It was capable of being powered from, or delivering to, a 42-V dc 20-Ah valve-regulated lead-acid battery. There was 3 kW of motoring capability and 3.5 kW of generation. The torque rating was 56 N · m (at 300 r/min), and the maximum speed was 15 000 r/min. This gave starting assistance to the ICE, energy saving during idle stop when ICE is purposely shut down, and partial regenerative electric braking of the vehicle (as much as the battery permitted). It also gave 2.1 kW of auxiliary drive (to the ac compressor with an idle ICE). It produced a 12.15% improvement in fuel economy in standard town driving. A basic system is shown in Fig. 13. The CPA is driven via a belt. While there are less performance data in [68], the results in terms of fuel economy are convincing. The technology is mature and was straightforward to implement. A larger version of this system has been recently developed in [24]. This gave 8–10 kW of regenerative braking without inverter control at 42-V dc.

In order to try to increase CPA efficiency, NdFeB (and later ferrite) PMs can be placed between the rotor claws (this is not a new idea, as illustrated in [70]). This will reduce the rotor dc field current leakage and decrease the rotor magnetic saturation level. If an inverter is used, then the field windings can be completely removed, thus removing the slip rings. In [24], both a field winding and PM excitation were used, and a 3-D nonlinear magnetic circuit design method was developed. Finite-element analysis was also used. The model is shown in Fig. 14. The design was optimized as a generator up to about 8 kW from 1500 to 6000 r/min and with PWM inverter control.

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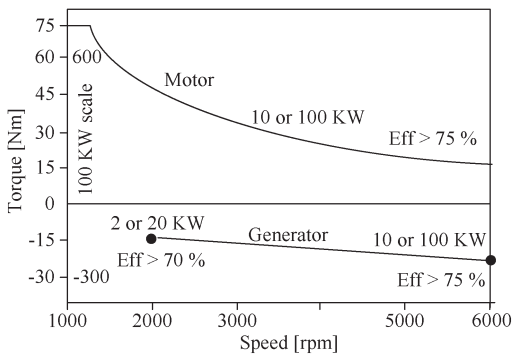

Fig. 15. Torque/speed requirements for 10- and 100-kW claw pole motor/generators in [25].

TABLE VI
CPA OF 100 kW

Parameter	CPA without PMs	CPA with NdFeB PMs	CPA with Ferrite PMs
Efficiency at 3000 rpm (generating) [%]	93.4	94.68	94.25
Power produced at 3000 rpm (generating) [kW]	105.2	108.5	159.3
Efficiency at 6000 rpm (generating) [%]	92.62	94.2	93.96
Power produced at 6000 rpm (motoring) [kW]	120.0	130.7	207.0
Active material weight [kg]	55.03	75.49	62.21
Active material cost [$]	176.5	284.6	197.0
Air-gap length [mm]	1.15	1	1.45
Total objective function [$]	394624	408456	394012
Stator outside diameter [mm]	395.5	389	388
Stator core length [mm]	36.5	51	47.5

This work is further developed in [25] for a machine up to 100 kW. The torque/speed requirements for the motor and generating modes at 10 and 100 kW are shown in Fig. 15. A summary of the results for the different designs are given in Table VI. What can be noted is that the diameter is reduced in the NdFeB and ferrite rotor machines; however, the axial lengths are increased. This leads to increased volume and gives increased power and efficiency. Each machine undergoes an optimized design procedure (hence the different dimensions), and it is interesting to note that the ferrite magnet rotor performs well compared with the NdFeB rotor. The demagnetization of the ferrite magnets is not addressed, which may be an issue.

It should be noted that, for the 16-pole ferrite magnet 100-kW machine, with full generator power at 3000 r/min and full motor power at 6000 r/min, the efficiency is very competitive (above 93%) at a reasonable weight (62 kg). It shows that there is only about two thirds of the active material cost in the ferrite CPA compared with the NdFeB CPA material. The optimal design was built and validated in [25] for compliance. A vector-controlled 60-kW ferrite PM machine was considered in the study. It was found that there were some power oscillations, but the power was stably regulated within 25 ms. It was concluded that the ferrite IPM claw pole machine has potential, particularly because the excitation losses are small. To remove the brush gear, a brushless rotary transformer can be utilized. The CPSR can be obtained in a straightforward

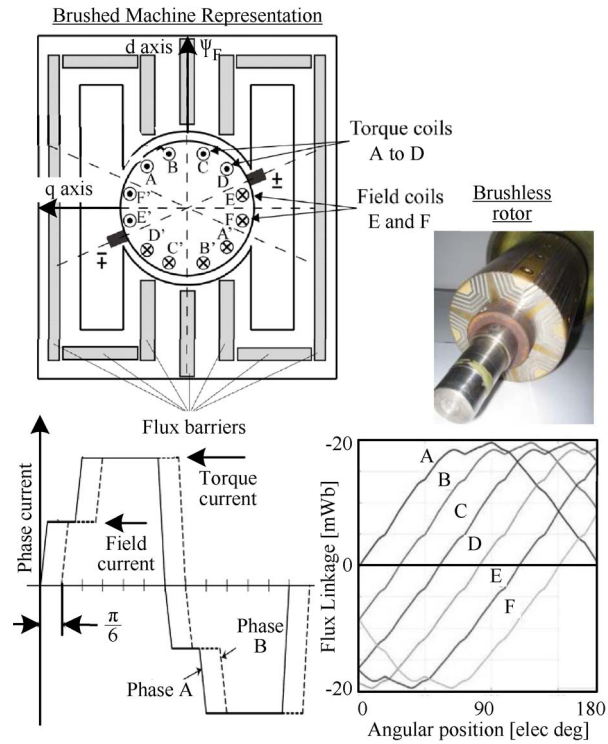

Fig. 16. (Top left) BLDC-MRM representation in terms of six-phase brushed reluctance machine and illustration of six-pole axially laminated ducted rotor. (Bottom left) Phase current cycle and (bottom right) phase flux-linkage waveforms for laboratory prototype (2-D finite-element analysis and measured) [27] (modified with permission from the authors).

manner by decreasing the field current, and this can be done with good efficiency. The no-load voltage at peak speed is less than the rated voltage; this is because the ferrite PMs are placed between rotor claws, and much of the flux crosses between the poles without linking the stator.

C. BLDC-MRM

This type of machine was studied in [27]–[29]. The BLDC-MRM is a development from a multiple-flux-barrier-stator BLDC machine. Unlike a normal dc machine, there is now field winding or PMs. This machine is the best way to explain the BLDC-MRM operation, and it is illustrated in Fig. 16. It has $2M$ slots, where M is the number of phases and coils per pole pair. The brushes are shifted from the neutral axis (see Fig. 16). A typical six-phase BLDC-MRM simply reverses the stator and rotor arrangement so that there is a ducted reluctance rotor and the phases are electronically switched. Therefore, this requires a six-phase inverter and position feedback to operate successfully. In the brushed arrangement in Fig. 16, two phases, i.e., E and F, are briefly "field" phases as they rotate round through the d-axis orientation. Phases A–D carry the "torque" currents. The current waveforms are illustrated in Fig. 17. This arrangement then generates a field flux so that excitation is not required. Hence, in terms of a brushed machine, it is equivalent to rotating the brushes.

In [27], a 2-D finite-element analysis was carried out. The calculated torque (and its torque/weight value) versus torque current density curves are shown in Fig. 15. These are for a

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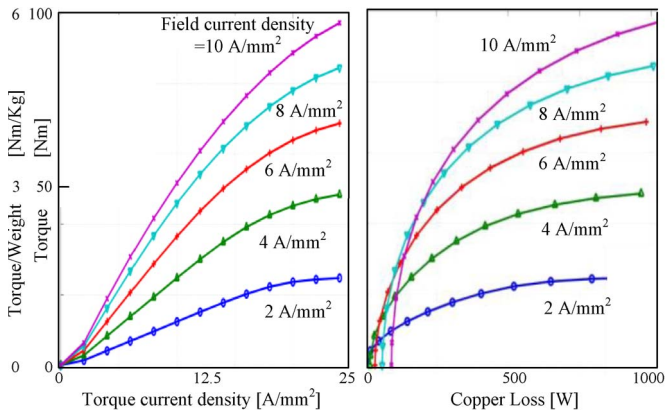

Fig. 17. BLDC-MRM torque and torque/weight for different loads and field excitations (modified with permission of the authors).

machine with a 104-mm inner stator diameter and a 145-mm stack length. With a field current density of 12 A/mm^2, it was found that 7 N · m/kg of active material was possible at a torque current density of 25 A/mm^2. When the field current and the torque current densities were equal at 10 A/mm^2, then the torque was 70 N · m (3.6 N · m/kg), and the copper loss is only 400 W, which, at 2000 r/min, would be around 3% of the output power, leading to very good efficiency. This is illustrated in Fig. 15.

The machine has 8 kg of active material per liter of volume; at 70 N · m, it has a torque density of 28 N · m/L. With a larger diameter and an optimized design, 45 N · m/L may be possible, with efficiency of 92%–94%. When the field and torque currents are equal, effective flux weakening is obtained, which will produce good efficiency at high speeds. With promising laboratory results based on close-to-optimum (around 90°) interaction between the field and torque phases and flat-top currents with simple flux weakening control, it appears that the BLDC-MRM should be investigated further. The number of phases can vary greatly; since multiphase inverters are available, then fault tolerance operation is a realistic proposition [71], [72]. Another advantage is that good kilovoltampere inverter utilization is inherent in this BLDC machine.

D. DC (or Ferrite PM) and AC Wound Stator Flux Switching Machine

This type of machine was studied in [30]–[32] and [73]. Starting from an SRM with double saliency, the dc field windings are formed from coils with two-slot pitch; alternate slots are used for the field windings, with two coil sides per slot. This configuration is shown in Fig. 18, and the arrows indicate the polarity of the currents in the dc field windings. In [30], the machine is described as a 12-slot, 10-pole machine. This actually means that the ac winding is in 12 slots, in a similar fashion to an SRM, as illustrated in Fig. 18. The key point here is to study the paths of the excitation flux. The machine can be divided into quarters. Consider the right-hand quarter of the cross section shown in Fig. 18. Rotor Pole 2 aligns with Stator Pole 2. The flux set up by the current in Excitation Coil 2 will cross this air gap because the reluctance is low and return to the stator through Rotor Poles 1 and 3 and Stator Poles 2

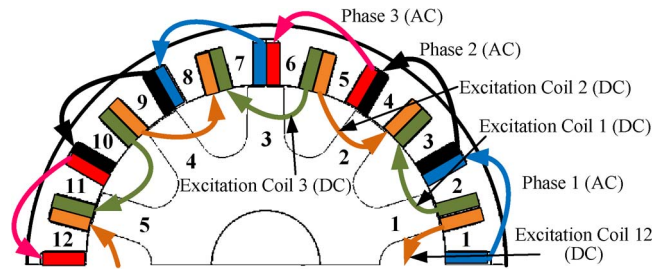

Fig. 18. DC and ac windings on field-excitation flux switching synchronous machine (FEFSSM) [30].

and 6. This flux also links Excitation Coils 1 and 3; their flux will add to the flux from Excitation Coil 2. In terms of the ac phases, the total flux from Excitation Coils 1, 2, and 3, which cross the air gap through Rotor Pole 2, will link Phase 2. If the rotor rotates backward (clockwise) by one stator slot in Fig. 18, then Rotor Pole 2 aligns with Stator Pole 3. Phase 2 is now linked by Excitation Coil 1 and by Excitation Coil 12; in addition, the flux direction is reversed. If the rotor rotates forward (counterclockwise) by one stator slot, then Phase 2 is still linked by Excitation Coil 2, but now linked by Excitation Coil 1. It is not linked by Excitation Coil 3 since Rotor Pole 3 is now between Stator Poles 7 and 8. Therefore, the flux linkage is reduced. It can now be seen that the excitation flux linkage with the phase windings will alternate as the rotor rotates.

For the left-hand quarter, rotor pole relative positions are offset by one stator slot, and this helps produce a more sinusoidal flux linkage. The phase back EMF for the machine shown in Fig. 18 was simulated in [30] and appeared to be reasonable and close to sinusoidal. This means that this machine will run with sinusoidal phase current. The instantaneous torque was also simulated, and it was found that the torque ripple was about 10% of the steady-load torque. Since the excitation is controlled by the dc excitation winding, then it is possible to control the phase currents to be on an effective q-axis. The inductance is relatively high; thus, the power factor is lower; hence, the inverter kilovoltampere utilization needs to be carefully assessed.

The torque/speed and power/speed characteristics are shown in Fig. 19 together with the assessment of the output power (as a percentage of input power, i.e., the efficiency) at various load points and the loss components. It is shown that the efficiency appears to reduce at high speeds, and this seems to be due to iron loss. However, good efficiency was found. The machine was designed as an alternative to the Toyota RX400H drive [74], and the results from the comparison of size and performance are given in Table VII. This is not quite a direct comparison because the speeds are different and higher gearing is needed. However, this is the same technique Toyota used to reduce magnet material in the third-generation Prius machine—the torque was reduced by increasing the speed.

At 5584 r/min and 210 N · m (the maximum simulated value), the power is 123 kW at a power factor of only 0.644. However, the torque density is 7.7 N · m/kg with efficiency of 92%. The high speed of the machine implies a high fundamental frequency. At 2000 r/min, this will be $20\,000 \times 10/60 =$ 3.3 kHz. This is a high frequency for a 123-kW inverter and a

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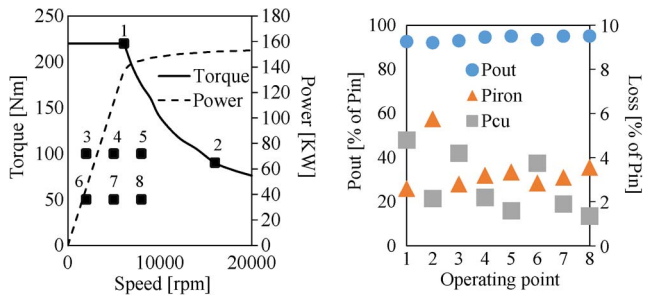

Fig. 19. FEFSSM [30]. (Left) Torque/speed envelope and (right) losses and efficiency at different load points.

TABLE VII
FEFSSM DESIGN RESTRICTIONS AND SPECIFICATIONS FOR HEV APPLICATIONS [30]

Parameter	IPMSM RX 400H [74]	FEFSSM [30]
Max. DC bus voltage [V]	650	650
Max. AC inverter current [Arms]	---	250
Max. armature current density [A/mm^2]	---	21
Max. excitation current density [A/mm^2]	---	21
Stator outer diameter [mm]	264	264
Motor stack length [mm]	70	70
Air-gap length [mm]	0.8	0.8
PM weight [kg]	1.1	0
Max. speed [rpm]	12400	20000
Max. torque [Nm]	333	> 210
Reduction gear ratio	2.48	4
Max. axle torque via reduction gear [Nm]	825	> 840
Max. power (kW)	123	> 123
Max power (kW/kg)	3.5	> 3.5

PWM switching frequency of at least 30 kHz, or a multilevel inverter will be required. The higher switching frequency implies large inverter commutation losses, which, together with high iron losses and the additional dc–dc converter, are major issues of this topology. However, high efficiency, torque density, and mechanical ruggedness may offset these issues.

In [31], a six-slot, eight-pole machine with field winding augmentation using PMs in parallel with the field winding was studied. This topology helps reduce the field current requirement. In [73], a theoretical study of a six-slot, five-pole machine was carried out. However, unbalanced magnetic pull (a radial decentering force) may occur in this machine since it appears to be diametrically unbalanced, which is not recommended in a large electrical machine. This arrangement will suffer from vibration and excessive bearing wear.

There are several variations of these flux switching machines. A further development of [31] is to remove the field winding altogether, and this was done in [32]. A two-phase configuration is illustrated in Fig. 20. It is shown that, as the rotor rotates (counterclockwise), a rotor pole is about to align with the left pole section of Phase C. The flux from the magnet in Phase C will flow through Interpole 1 into the rotor and then return through the rotor pole aligned with Phase C. Interpole 1 will also carry flux from the Phase A magnet in the same direction. If the rotor is moved back so the rotor pole aligns with the right-hand pole section of Phase C, then the flux linkage of the ac winding will be reversed because the magnet flux now flows through Interpole 2, which is now aligned with a rotor pole. Hence, alternating flux linkage of the phase windings occurs.

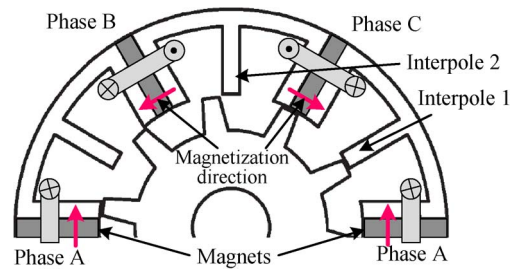

Fig. 20. E-core flux-switched 12/10 double-saliency machine with stator ferrite PMs [32].

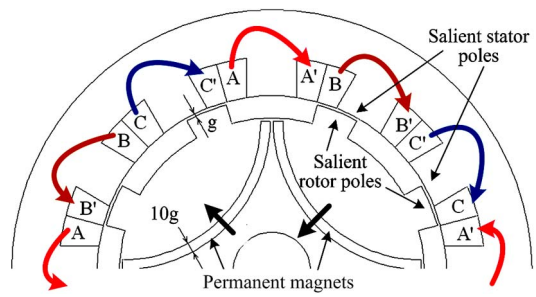

Fig. 21. The 12/8 ferrite PM rotor flux-switched double-saliency machine [33].

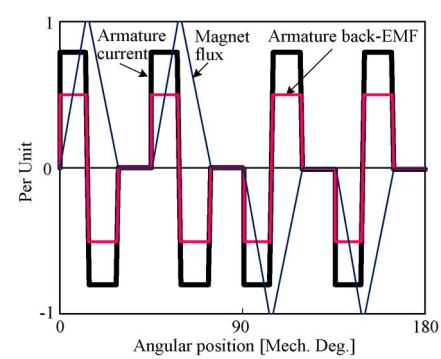

Fig. 22. Flux, bipolar current, and EMF for the 12/8 ferrite PM rotor flux-switched double-saliency machine [33].

This topology may produce sufficient flux concentration to produce a torque density up to the desired 40–45 N · m/L and close to 93% efficiency [32]. The machine may still suffer from low power factor and higher iron losses, but this machine is being pursued by several researchers for a variety of applications, and typical work is given in [75] and [76]. A review of the different flux switching topologies was given in [77]. It has a large amount of ferrite magnet material.

E. DSPMM

Another flux switching machine can be considered as being derived from a BLDC machine. This has an SRM rotor with PMs. It has a 12/8 slot/pole combination, as illustrated in Fig. 21, and is called a double-saliency PM machine (DSPMM). This topology has ferrite magnets. The waveforms for the phase back EMF and current are shown in Fig. 22, and it is shown that the summation of the three phase currents is not zero; thus, an additional inverter leg is needed for independent phase current control. If there is sufficient magnet flux concentration, then

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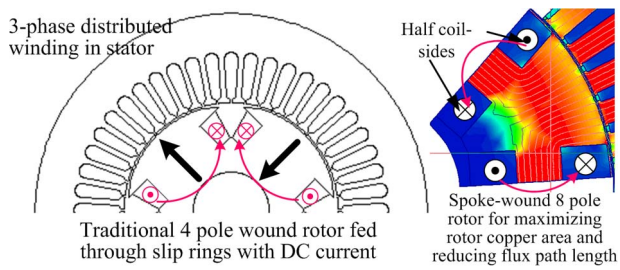

Fig. 23. Conventional (left) four-pole wound rotor synchronous machine and (right) eight-pole spoke-type rotor [62].

a high torque density may be obtained (45 N · m/L), whereas demagnetization of the PMs is avoided under peak load. Current waveform shaping for torque pulsation reduction is feasible, whereas the machine inductance is smaller than in a standard SRM so that faster commutation is feasible. The high torque density claimed for IPMSMs [1] may be met by this machine using ferrite PMs with further design work.

F. Wound Rotor Synchronous Machine

This is an old technology, and it has been applied to automotive applications in an experimental context [34], [35] and a design exercise [62]. The main issue with a traditional design is that it requires slip rings and has rotor excitation that produces losses. Even with a modern IPMSM design, cooling is critical, with added rotor losses, then the machine power density will have to be reduced to accommodate the losses. This machine could be included as a production motor because [78] did report that they have been used by Renault in EVs, but no other manufacturer appears to be following this line.

Here, some more unusual electric machine topologies have been addressed, and these either have no magnets, weaker ferrite magnets, or a reduced amount of rare-earth magnet. Some do come close to the performance of production IPMSMs used in HEVs and EVs. However, given the wide CPSR that is required (around 4), the dc-excited wound rotor synchronous machine with higher pole number (probably between 6 and 12 poles) may as well be suitable due to the following.

- q-axis current operation is possible (for maximum torque/current), and good current control at low speeds can produce high torque density (as high as other machines since the current is in phase with the back EMF).
- The reduction of the d-axis stator current to zero, as with the BEGA machine, and with full field current to obtain peak torque, leads to about 30% reduction in the peak inverter kilovoltampere rating; this gains only a theoretical maximum, and it is offset by the fact that 5% of the power is lost through the field current dc–dc converter and field losses.

A four-pole wound rotor synchronous machine drive for a forklift–car propulsion system was developed in [34]. Fig. 23 illustrates the salient pole nature of the machine. It was very overrated (up to 4 p.u of the steady-state torque when it was at 300 N · m) at very low speeds, as shown in Fig. 24. However, the thermal duty cycling of EV and HEV drive motors is very important—they can usually operate for short times at very high

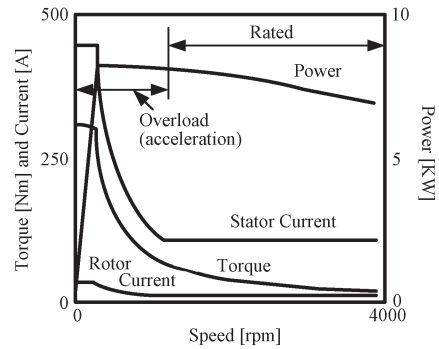

Fig. 24. Representation of measured results in [34] for a dc-excited synchronous machine: 54 V, 110 A, four pole, 1010-r/min base speed, 70-N · m base torque, 95% efficiency.

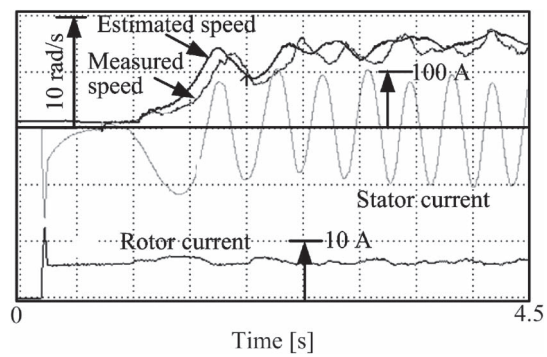

Fig. 25. Initial rotor position estimation and sensorless startup from 0 to 80 r/min and 100% base torque load by friction from zero speed [35] (modified with permission from the authors).

torques with good fluid cooling. Fig. 24 illustrates that it has a wide field weakening region, which is probably a major reason for Renault for choosing this arrangement.

One of the main problems with a synchronous machine is getting sufficient winding on the rotor to obtain a high field while at the same time restricting copper losses; rotor cooling is more difficult than stator cooling. In [62], a spoked type of winding was used in a design exercise to maximize the copper area available. The results suggest that the rotor copper losses are considerably higher than the stator copper losses if not properly designed and operated up to a peak load. This illustrates that moving field excitation to the stator, where cooling can be more effective, is a reasonable design choice.

For EV and HEV operation, good control is required. An active flux-based sensorless vector control for a wound rotor synchronous machine was developed in [35]. This included initial position estimation but without signal injection. The operation is shown in Fig. 25, and it is found to produce safe starting under full (base) torque conditions.

If the slip rings are not required, then a brushless rotary transformer could be used. This would probably utilize a 500-Hz inverter supply and a diode rectifier. The high frequency would reduce the size of the transformer.

G. Comparison of Performance

Many of the machines reported in this paper are proof-of-concept machines; however, it is worth noting some of

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TABLE VIII
COMPARISON OF MACHINES

Machine	Max Power [kW]	Max. power /vol. [kW/L]	Max. Torque [Nm]	Max torque /vol. [Nm/L]	Max. speed [rpm]	Knee point Eff. [%]	Rare earth magnet?
Prius 2nd gen. [12]	50	10.5	400	83.7	6000	85* 91.3**	Yes
Prius 3rd gen. [13]	50	18.2	207	75.5	13900	88*	Yes
Air-cooled IPM [11]	35	9.3	70	18.7	12000	90*	Yes
Induction motor [12]	47.5	9.975	297	57.2	6000	83.1**	No
SRM [12]	47.4	9.95	294	55.5	6000	85.2**	No
SRM [16]	60	6.8	340	38.5	14000	90**	No
SRM [19]	50	5.6	400.4	45	14000	90**	No
DSSRM-1 [10]	29	4.6	137	21.9	6000	88**	No
DSSRM-2 [10]	50	19.2	113	43.5	10000	93**	No
Ferrite sync. rel. [21]	42	22	120	63.5	10000	85**	No
BEGA [23]	2.35	N/G	6	N/G	2000	N/G	No
Claw pole alternator [68]	3	N/G	56	N/G	15000	N/G	No
CPA [28]	100	22.3	740	165	6000	>70**	No
CPA with NdFeB [25]	100	16.5	740	122	6000	>70**	Yes
CPA with ferrite [25]	100	17.8	740	132	6000	>70**	No
BLDC Reluctance [27]	N/G	N/G	70	28	N/G	N/G	No
BLDC Reluctance [29]	100	N/G	600	N/G	N/G	90**	No
Flux switching [30]	>123	32	>210	54.8	20000	92**	No

* Overall; ** Electromagnetic efficiency; N/G - not given

the basic operational parameters (see Table VIII). Many are theoretical designs and not subject to a full thermal analysis; thus, the power and torque densities need to be considered with caution. For the power density, it can be seen that this generally increases with speed as expected. The Prius machines give good torque density, and these still represent good benchmarking machines.

H. Inverter Drives

Most of the machines addressed here use a fully rated three-phase inverter. The focus of this paper is not the inverter; however, when investigating a machine, the drive requirements need to be addressed in terms of current waveform (sine, trapezoidal bipolar, or trapezoidal unipolar). In addition, some of the machines have high pole number or are flux switching, and these can have a high fundamental frequency. This can affect the power factor, which can lead to high current requirement and a high demand on the PWM control.

VI. CONCLUSION AND FUTURE TRENDS

This paper has attempted to review the current technology in HEV and EV drive motors with a focus on machines that have reduced magnet content or no magnets at all. Currently, the IPMSM is the most common machine in use, but manufacturers are keen to develop drive motors with much lower magnet content. Current motor technologies have been reviewed (the IPMSM, the IM, and the SRM). The key issue with magnetless and reduced-magnet machines is the power density and the efficiency. While brushed dc and wound rotor synchronous machines are traditional drives, indeed they have been successfully used as train traction motors for many decades, they struggle to meet the power density requirement. Hence, brushed dc machines are not addressed here, and wound rotor synchronous machines are addressed only in terms of research applications. They have been used in EVs [62] but now tend to be limited in their applications due to lower power density and efficiency. While [78] listed many commercial EV drive arrangements, it has to be stated that there are many variations of the four classifications they give: PM, IM, dc, and synchronous brushed. It is also sometimes difficult to assess the type of machine used in a vehicle from the brief specifications. For instance, ac motor, dc motor, or reluctance motor could cover a variety of machines.

The machines reviewed here have very different topologies, but they all have less PM material or none at all, and they have PWM converter control. They are capable of producing the torque–speed characteristics that are close to the NdFeB IPMSM. Surface magnet brushless machines have not been investigated here, although they are used in production and have been studied in the research arena [79] for HEVs. High torque density and efficiency are required, as well as a wide constant power range. Standard SRMs have been investigated over an extended range [80], and this is necessary for EVs and HEVs.

Each of the machines described here has potential for the application. Some of the machines have been discussed in terms of the PWM inverter kilovoltampere requirements, as well as experimental validation of the machine, although some of the studies are design exercises. It should be borne in mind that the PWM converter is often still more expensive than the electric machine itself, and it is still more than 20 US$/kVA today [4]. Studies are needed that address the integration of the system (machine, converter, control, and energy storage system) for each of the machines, and these should include cost and performance, thermal performance, and duty cycling.

There is still more work to be done; when designing machines, it is becoming important to design and assess performance over a complete speed and load range. There are several motor design packages now being developed that allow flexible design; and development of efficiency charts over complete torque/speed ranges [81], [82] is important. Duty cycling is key in the thermal design, motor control, and energy management of automotive drives [83]–[88]; many of the prototype designs described here have not fully addressed these issues, although the benchmarking results look positive.

The motors here are designed to operate at higher speeds as part of a central drive unit supplying power to two or four wheels. An alternative is to use smaller motors that are in-wheel [84] or direct drive to one wheel [89]. These are more suitable for EVs and present different challenges. Most of the work here is in the context of motoring, although the BEGA is aimed at generation. With in-wheel drives, the machine works both as a motor and a regenerative brake, and individual wheel drives lead to the possibilities of functions such as electronic antilock braking regeneration [90] and electrical differential.

The Toyota third-generation drive reduced the amount of magnet material by essentially increasing the speed. In a similar manner, the performance of surface magnet fractional-slot machines was addressed in [91] in terms of the amount of magnet used; this is of great interest to the automotive

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industry. In [33], ferrite magnets were considered in a more novel machine. These are cheaper, and if weaker magnetization is to be assessed at higher speed, then ferrite magnets could be utilized. In [92], a more standard magnetic spoke rotor machine was simulated, and it was illustrated that this has much lower torque density than the equivalent rare-earth magnet rotor, even with much more magnet material.

When SRMs are utilized, they are well known to generate noise and vibration due to their double saliency. The double-sided machine in [20] addressed this issue; in more standard SRMs, it is possible to do complete mechanical/electromagnetic studies to reduce this noise [93]. Again, this type of work is important to industry when designing products for market.

To conclude, when designing the vehicle drive motor, the sizing and operation have to be integrated into a complete vehicle drive and energy system, and increasingly, optimization is being used to assess these systems [94], [95]. The drive motor is one component of the system, and it needs to be fully characterized to allow accurate assessment in the system.

Variations of systems are creeping into the marketplace. For instance, a form of mild hybridization is now available on Mazda 6 ("i-ELOOP") [96]. This is a supercapacitor type of system that takes regenerated energy from braking via the alternator (at 12 V), and it claims a 10% improvement in fuel efficiency under urban cycling.

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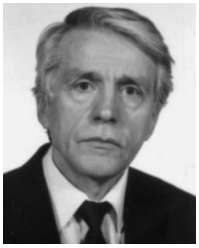

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